

RESEARCH ARTICLE

In-Vehicle CSI Gesture Recognition Using Two Wi-Fi Receivers With FEDRT-LSTM Model

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ABSTRACT Existing in-vehicle gesture recognition using channel state information (CSI) of Wi-Fi signals requires the use of 3 or more receivers. Due to the lack of research on gesture recognition using fewer receivers, we here study the case of only two receivers. CSI data of 4 gestures are collected by two in-vehicle receivers and are then processed to generate body velocity profile (BVP) data. We propose a Transformer and long short-term memory (LSTM) (Transformer-LSTM) fusion model with a feature extraction and dimensionality reduction module (FEDRT-LSTM) to recognize gestures using BVP data from two-receiver CSI. The performance is compared with those of deep learning neural network (DNN), gated recurrent unit (GRU) capsules network (GRU-CapNet), LeNet-5, and GRU network models with hyperparameters optimized. Results show that the FEDRT-LSTM model can achieve the best recognition accuracy of 95.54%. This approach can reduce the number of receivers to two and achieve good performance in the recognition of 4 gestures.

INDEX TERMS In-vehicle, CSI, Transformer, LSTM, hyperparameter optimization.

I. INTRODUCTION

With the rapid development of Wi-Fi sensing technology [1], [2], [3], [4], [5], [6], indoor positioning and user action recognition [7], [8], [9] using channel state information (CSI) [10], [11], [12] have also become of greater research interest. The use of non-contact gesture recognition technology in in-vehicle multimedia applications is becoming increasingly important for improved safety and convenience. However, studies on CSI-based driver gesture recognition are quite limited. As far as we are aware of, there are no related publications on in-vehicle CSI gesture recognition using only two receivers. So far, there are not many open datasets for CSI-based gesture recognition. Widar3.0 dataset collects indoor gesture CSI [13], [14] from 6 receivers. MM-Fi dataset collects indoor gesture CSI [15] from one receiver. However, the indoor and in-vehicle environments are different, with the

latter being more challenging to collect data in and suffering more significantly from interferences from other commonly used wireless devices and technologies like Bluetooth, increasing the difficulty of CSI signal processing [16], [17]. The self-built dataset with gated recurrent unit (GRU) capsules network (GRU-CapNet) in [18] collects in-vehicle gesture CSI using 3 receivers. Reducing the number of in-vehicle Wi-Fi receivers leads to less CSI information, and is expected to result in a decrease in gesture recognition accuracy.

After processing CSI data using signal processing techniques, neural networks are used for gesture recognition. The choice of the appropriate neural network model is particularly important for recognition accuracy. Widar3.0 dataset is first recognized using the DNN model [19], later by Wang Chi using the 3D convolutional neural network model (3D-CNN) [20], [21]. Zhang Meng further proposes the GRU-CapNet model [18]. Although the accuracy of GRU-CapNet on his self-built in-vehicle gesture dataset with 3 receivers is

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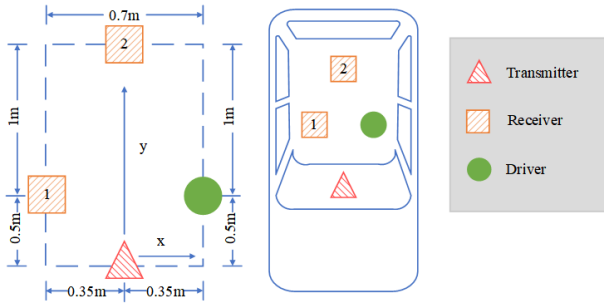


FIGURE 1. Experiment environment.

97.95%, we find that the accuracy drops significantly when GRU-CapNet is applied to 2 receivers. Therefore, the gesture recognition accuracy of these models on in-vehicle gesture dataset using two Wi-Fi receivers is found to be insufficient, as will be shown in Section V.

In this paper, we collect our in-vehicle gesture CSI data from two receivers and propose to add a feature extraction and dimensionality reduction module (FEDR) to Transformer-LSTM [22], [23], [24], i.e. FEDRT-LSTM. Then, FEDRT-LSTM is optimized for 4 hyperparameters: loss function, epoch, learning rate, and batch size. After obtaining the optimal parameter combination for the model, comparison with DNN [25], GRU-CapNet, GRU [26], and LeNet-5 [27] as baseline models, the FEDRT-LSTM model performs best.

II. DATA

A. CSI SIGNAL COLLECTION

The experiment environment is a commercial Volkswagen Bora, parked on campus with a driver and 3 passengers, with 3 Thinkpad X250 laptops, one of which is a sender and the other two are receivers. The 3 laptops are equipped with Intel5300 network cards and collect CSI data on Ubuntu 14.04 using Linux CSI Tool [28]. Considering the obstruction of the seat backs to signal transmission while there is no seat back blocking in between the two rear seats, our experiment layout differs from the GRU-CapNet experiment [18] in that the two receivers in the rear seat are reduced to one and placed in between the two rear seats. The layout of the experiment environment in the vehicle is shown in Figure 1, and the experiment process is shown in Figure 2. The experiment process uses Monitor mode to collect data at a rate of 5000 packets per second. However, due to the large volume of data, the amplitude variations are not significant, so the data is downsampled to 1000 packets per second during processing.

Fourteen volunteers participate in the experiment. Each volunteer performs 4 gestures: Sweep, Clap, Draw Z, and Draw O, shown in Figure 3. The sending device displays on the screen the category and index of the gesture to perform and plays a prompt sound, like a beep. This helps the volunteer confirm the timing of the gestures. Upon hearing the prompt, the volunteer sitting in the driver's seat begins to make gestures. After two seconds, the receiving devices receive the CSI signals of the current gesture and store them in .dat files. Two supervisors of the experiment acting



FIGURE 2. Experiment process.

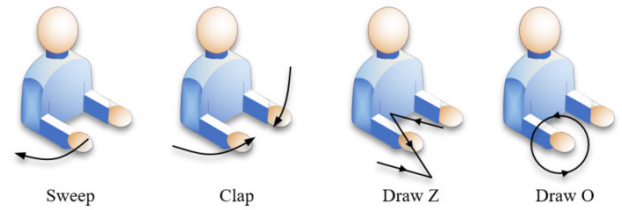


FIGURE 3. Hand gestures.

as passengers at the two rear seats verify the attributes of each .dat file on the two receiving devices. The number of files in the folder of each receiving device is checked, and the number of CSI signals is gradually increasing. The files are named in the format of gesture category-execution index, separated by hyphens. The file names and quantities show whether there are duplicates or omissions in the data collection process. If any issue occurs, the supervisors manually pause the sending device from transmitting signals. Each gesture lasts two seconds and is repeated 20 times by each volunteer. In total, each receiver successfully receives a total of $20 \times 14 \times 4$, i.e. 1120 CSI data, and the two receiving devices store a total of 2240 CSI data files. The number of antennas at the transmitter is one, the number at the receiver is 3, and the number of subcarriers captured by CSI Tool is 30. Thus each CSI data file contains a $1 \times 3 \times 30$ matrix. An example of CSI amplitude versus time in one data file is shown in Figure 4.

B. CSI SIGNAL PROCESSING

Due to the impact of interferences from other wireless devices and noise on the signal waveform, the raw CSI contains static components, low-frequency interference, and noise that can affect the Doppler frequency shift (DFS). To address this issue, we employ the signal processing procedure employed in WiDance [29]. The research of Widar3.0 starting with WiDance in 2017 includes indoor gesture recognition based on 6 Wi-Fi receivers. Subsequent works like Widar in 2017, Widar2.0 in 2018, and Widar3.0 in 2019 all applied the same signal processing methods in WiDance. The homepage of Widar 3.0 provides the original indoor CSI signals, signal processing, and neural network program codes. GRU-CapNet in [18] also uses the same signal processing methods from WiDance, in an in-vehicle environment using 3 Wi-Fi

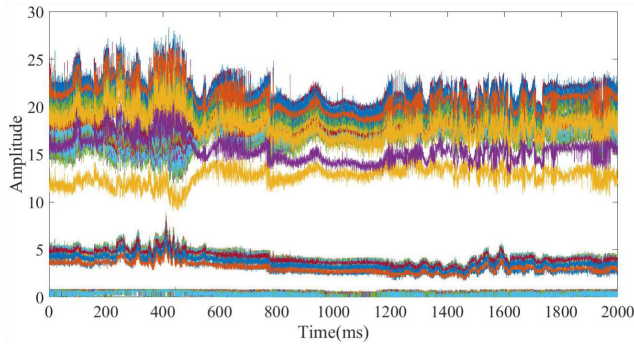


FIGURE 4. An example of Receiver 1's CSI amplitude of 3 antennas each with 30 subcarriers.

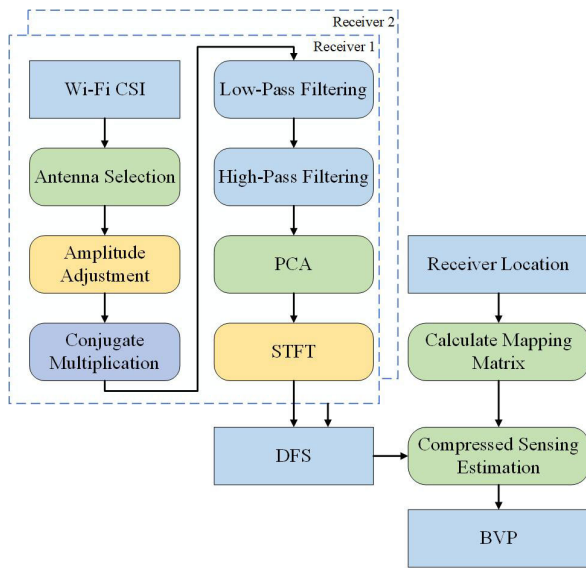


FIGURE 5. Signal processing flowchart.

receivers. This paper, based on the experience from repeating Widar3.0 and GRU-CapNet, focuses on gesture recognition in an in-vehicle environment using two Wi-Fi receivers and adopts the same signal processing methods in WiDance. Figure 5 illustrates a schematic diagram of the entire Wi-Fi CSI signal processing process, which consists of two parts.

The first part involves antenna selection, amplitude adjustment, conjugate multiplication [30], high-pass and low-pass filtering to remove static and high-frequency components, principal component analysis (PCA), and short-term Fourier transform (STFT), resulting in a DFS. The second part involves obtaining the receiver location [31], [32], [33], [34], [35], [36], [37], calculating the mapping matrix, and using compressed sensing estimation, resulting in a body-coordinate velocity profile (BVP). An example of BVP data in image form for sequence length 20 is shown in Figure 6. Each BVP data is a three-dimensional tensor of velocity resolution $\times$ velocity resolution $\times$ sequence length, where sequence length is the dimension in time, similar to consecutive frames in a video, which shows the power distribution of velocities at a given time.

CSI signals from two receivers are merged and stored in one .dat file. The file naming convention includes the user

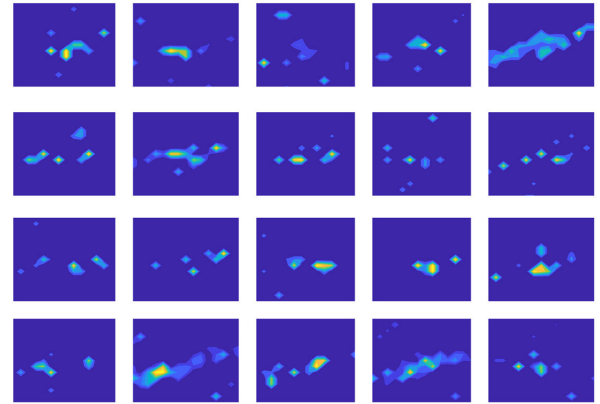


FIGURE 6. BVP images for sequence length 20.

index, gesture category index, trial index, and receiver index separated by hyphens. For example, the file u14-g4-t20-r2.dat represents the 20th trial of the 4th gesture performed by the 14th user. BVP data is stored in .mat files. The u14-g4-t20.mat file is generated by calculating u14-g4-t20-r1.dat and u14-g4-t20-r2.dat, allowing for easy management and identification of files in the dataset based on the file names.

There are a total of 1120 BVP data in our dataset. During each training process, they are randomly divided into training set and test set in a 9:1 ratio, with 1008 samples in the training set and 112 samples in the test set.

III. FEDRT-LSTM MODEL

When the number of receivers is reduced from 3 to 2, we find that the recognition accuracy of the DNN model in [19] decreases from 92.7% to 88.39%, while the recognition accuracy of the GRU-CapNet model [18] declines from 97.95% to 84.82%. Hence, these models are unable to deliver desirable gesture recognition performance using signals from two receivers, demanding further improvement in recognition algorithms.

To better capture the spatiotemporal characteristics of the BVP data, we propose to add a new feature extraction and dimensionality reduction module to the Transformer-LSTM model in [23], [24]. The model structure as shown in Figure 7 consists of the following 4 parts.

A. FEATURE EXTRACTION AND DIMENSIONALITY REDUCTION

GRU-CapNet in [18] achieved a recognition accuracy of 97.95% on its self-built in-vehicle gesture dataset. Since the experiment environment of our work is also an in-vehicle environment, the input size of the neural network model in this paper directly follows that of GRU-CapNet, i.e. a velocity resolution of 15 and a sequence length of 20. The size of the input gesture BVP data is thus $15 \times 15 \times 20$. Firstly, a 2D convolutional operation is performed using 16 5×5 convolution kernels without using zero padding, using ReLU as the activation function, and the output size is $11 \times 11 \times 16$. Next, the output from the previous layer goes through a maximum pooling operation with a pooling window of 2×2 , and the rest of the parameters are the same as those of

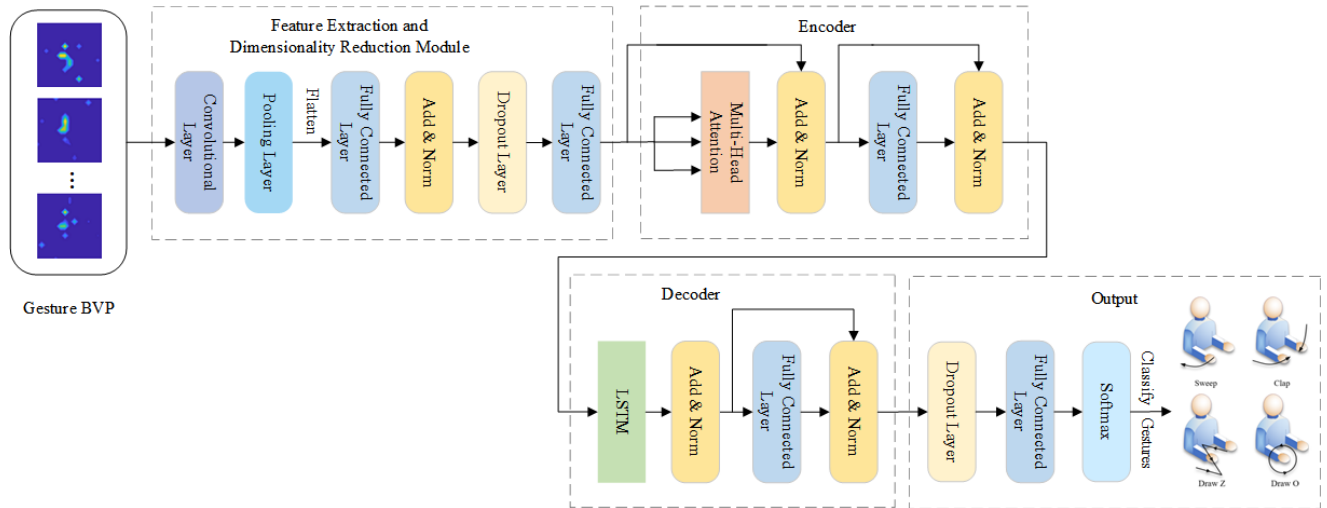


FIGURE 7. FEDRT-LSTM structure.

the previous layer. As $11/2=5.5$, is rounded down to 5, the output size is $5 \times 5 \times 16$. Subsequently, the output from the previous layer is flattened to a one-dimensional vector, so the output size is $5 \times 5 \times 16=400$. Then two consecutive fully connected layers containing 64 neurons are applied to all sequence lengths to ensure that the model takes into account the dynamic nature of the time series in processing each frame and the output size is 64. Note that one extra dropout layer is added between the two fully connected layers to reduce overfitting, with the output size remaining unchanged.

B. TRANSFORMER

The Transformer model uses a position-coding strategy to obtain relative position [38] information in the input sequence and uses a multi-head attention mechanism to pay attention to detailed information in different subspaces. The Transformer layer designed in this study consists of an encoder and a decoder. The encoder contains a multi-head attention layer (Multi-HeadAttention), a regularization layer (Add&Norm), a fully connected layer (FFN), and an Add&Norm. The self-attention mechanism weighs and combines different parts of the input data by calculating the attention weight of the input data. Specifically, the multi-head attention mechanism used here has 8 heads to calculate self-attention. The decoder is composed of an Add&Norm, a fully connected, and an Add&Norm. The activation function is ReLU, and the output size remains unchanged, i.e. 64.

C. LSTM

The LSTM model is a variant of recurrent neural network (RNN) that can better capture long-term dependencies in time series, thereby improving the accuracy of the model. Current research shows that this structure has been incorporated into different research fields such as language modeling and text generation [23]. An LSTM layer is inserted into the traditional Transformer decoder to better capture the patterns

in the BVP sequence. Our design uses an LSTM layer with 8 hidden units to convert the feature sequence into a fixed-length feature vector.

D. OUTPUT

The output layer uses dropout to achieve the regularization effect and effectively alleviate the occurrence of over-fitting, followed by the fully connected layer. The activation function is softmax. The output gesture has 4 categories.

IV. HYPERPARAMETER OPTIMIZATION

The GPU is Tesla T4, and the training environment is CUDA 12.0, TensorFlow 2.1, and Python 3.7. FEDRT-LSTM is implemented using the Keras library. Due to the influence of hyperparameter values on the performance of neural network models, the subsequent discussion examines the effects of 4 varying hyperparameters on the accuracy and loss of the FEDRT-LSTM network model, namely the loss function, epoch, learning rate, and batch size. The value corresponding to the highest accuracy is selected for each hyperparameter. Finally, the optimal combination of hyperparameters is obtained for the recognition task using the FEDRT-LSTM network model with in-vehicle CSI gesture data collected using two receivers.

A. IMPACT OF THE LOSS FUNCTION ON RECOGNITION ACCURACY

The primary aim of neural network training is to minimize the loss function by optimizing the network's parameters or weights. This process involves adjusting the weights to align the predicted values with the true values. The role of the loss function in parameter optimization is crucial as it quantifies the discrepancy between predicted and true values. By employing the gradient descent method, the network continually updates its weights to minimize the loss function. In this particular section, four distinct loss functions have been chosen to evaluate their impact on recognition accuracy,

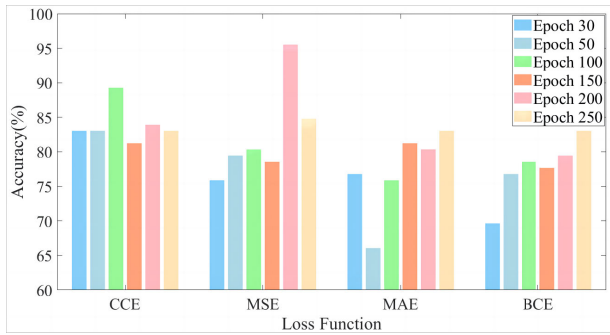


FIGURE 8. By comparing the accuracy of different loss functions at the same epochs, the best loss function is selected to optimize the performance of gesture recognition tasks.

i.e. categorical cross-entropy (CCE), mean squared error (MSE), mean absolute error (MAE), and binary cross-entropy (BCE). In the Widar3.0 framework, the selected loss function for the DNN model is CCE, with an epoch of 30, a learning rate of 0.001, and a batch size of 32. For analysis in this subsection, the same learning rate and batch size used in the DNN model are employed.

As shown in Figure 8, when the value of epoch is set to 30, 50, and 100, the gesture recognition accuracy is highest when using the CCE loss function. When the epoch value is set to 150, the gesture recognition accuracy is comparable between the CCE and MAE. However, when the epoch value is set to 200 and 250, the gesture recognition accuracy is highest when using the MSE loss function. Particularly, when the epoch is set to 200 and the MSE loss function is used, the gesture recognition accuracy reaches 95.54%, which is the highest among all 24 scenarios. Therefore, in the context of using two receivers to collect in-vehicle CSI gesture data and the FEDRT-LSTM network model, the optimal choice for the loss function is MSE. Next, further investigation will be conducted to evaluate the impact of varying epoch values on gesture recognition accuracy, to determine the best epoch value.

B. IMPACT OF EPOCH ON RECOGNITION ACCURACY

As shown in Figure 9, the gesture recognition accuracy shows a general upward trend as the value of epoch increases from 30 to 250. However, the highest accuracy does not correspond to an epoch value of 250, but rather 200, when the loss function is MSE. From the graph, it can be seen that when the epoch value increases from 150 to 200, the gesture recognition accuracy increases from 78.57% to 95.54%, i.e. an increase of 16.97%. However, increasing the epoch value further does not continue to increase the gesture recognition accuracy. Instead, it decreases compared to the accuracy at an epoch value of 200, but it is still higher than the accuracy corresponding to other epoch values.

The red line with CCE as the loss function shows the process of gesture recognition accuracy changes relatively steadily as the value of epoch increases from 30 to 250. Specifically, when the epoch value increases from 50 to 100,

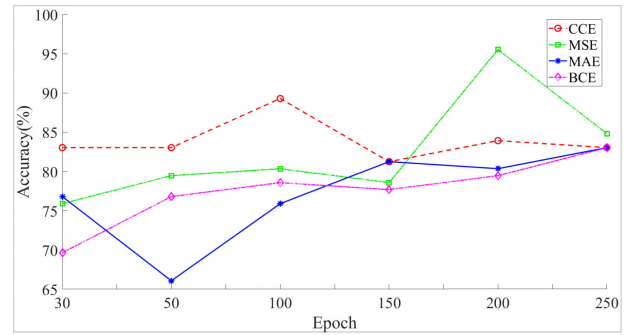


FIGURE 9. By comparing the accuracy of different epochs at the same loss functions, the best epoch is selected to optimize the performance of gesture recognition tasks.

the accuracy increases from 83.04% to 89.29%, i.e. an increase of 6.25%. The highest recognition accuracy is achieved at an epoch value of 100. When the epoch value increases from 150 to 200, the accuracy increases from 81.25% to 83.04%, i.e. an increase of 1.79%.

The blue line with MAE as the loss function shows the process of gesture recognition accuracy decreasing and then gradually increasing as the value of the epoch increases from 30 to 250. Specifically, when the epoch value increases from 30 to 50, the accuracy decreases from 76.79% to 66.07%. When the epoch value increases from 50 to 250, the accuracy increases from 66.07% to 83.04%. The accuracy at epoch value 250 is also higher than that at epoch value 30, achieving the highest recognition accuracy. The experiment results also indicate that increasing the value of the epoch can improve the value of gesture recognition accuracy.

The pink line with BCE as the loss function shows that the accuracy of gesture recognition generally increases when increasing from 30 to 250 epochs. When the epoch value increases from 100 to 150, the accuracy slightly decreases from 78.57% to 77.68%, but it does not affect the overall upward trend. The accuracy at epoch value 250 is also significantly higher than that at epoch value 30, achieving the highest recognition accuracy.

In conclusion, the highest accuracy of gesture recognition is achieved when the value of epoch is 200 and the MSE loss function is used. The accuracy of gesture recognition reaches 95.54%, which is the highest among all 24 cases. Therefore, when using the CSI data collected by two receivers and the FEDRT-LSTM network model, the optimal choice for the loss function is MSE and the optimal value for epoch is 200.

C. IMPACT OF LEARNING RATE ON RECOGNITION ACCURACY

As shown in Figure 10, as the value of the learning rate increases from 0.001 to 0.009, the accuracy of gesture recognition shows a general decreasing trend. The highest accuracy corresponds to a learning rate value of 0.001, with a batch size of 32. From the graph, it can be observed that when the batch size value is 32, the green line decreases from 95.54% to 82.14% as the learning rate value increases

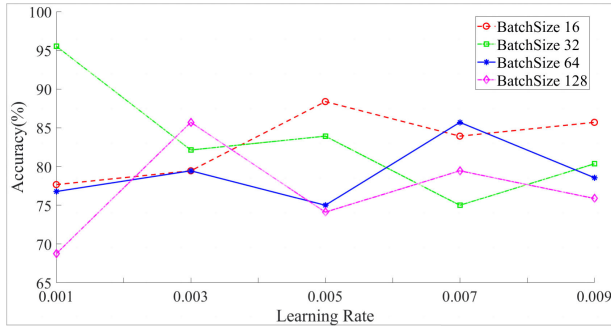


FIGURE 10. By comparing the accuracy of different learning rates at the same batch size, the best learning rate is selected to optimize the performance of gesture recognition tasks.

from 0.001 to 0.003, resulting in a decrease in accuracy of 13.4%. Furthermore, increasing the learning rate value leads to an increase in gesture recognition accuracy twice, when the learning rate increases from 0.003 to 0.005 and from 0.007 to 0.009, respectively. However, both cases have lower accuracy compared to when the learning rate value is 0.001.

In conclusion, the highest accuracy for gesture recognition is achieved when the value of the learning rate is 0.001 and the value of batch size is 32. The accuracy rate for gesture recognition reaches 95.54%, which is the highest among all 24 scenarios. Therefore, when using the CSI data collected by two receivers and the FEDRT-LSTM model, the optimal choice for loss function is MSE, the optimal epoch value is 200, and the optimal learning rate value is 0.001.

D. IMPACT OF BATCH SIZE ON RECOGNITION ACCURACY

As shown in Figure 11, as the value of batch size gradually increases from 16 to 128, the accuracy of gesture recognition shows a general downward trend. The highest accuracy corresponds to a batch size value of 32, with a learning rate value of 0.001. From the graph, it can be seen that the red line with a learning rate value of 0.001 has a significant change in gesture recognition accuracy as the batch size value increases from 16 to 128, but overall it is also decreasing. Especially when the batch size value increases from 16 to 32, the gesture recognition accuracy increases from 77.68% to 95.54%, an increase of 17.86%. However, as the batch size value continues to increase, the gesture recognition accuracy keeps decreasing. Particularly, when the batch size value increases from 32 to 64, the gesture recognition accuracy decreases from 95.54% to 76.79%, a decrease of 18.75%, which is 0.89% more than the previous increase. Therefore, overall it is decreasing. The blue line with a learning rate value of 0.005 shows a gradual decrease in gesture recognition accuracy as the batch size value increases from 16 to 128. Therefore, the highest gesture recognition accuracy is not at a batch size value of 32 but at a batch size value of 16. However, the gesture recognition accuracy in both cases is 88.39% and 83.93%, respectively, with a difference of only 4.46%. The gesture recognition accuracy at a batch size value of 32 is second only to the gesture recognition accuracy at a batch size value of 16.

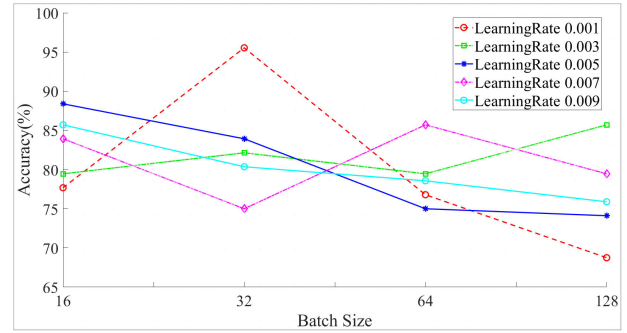


FIGURE 11. By comparing the accuracy of different batch sizes under the same learning rate, the optimal batch size value is selected to optimize the performance of the gesture recognition task.

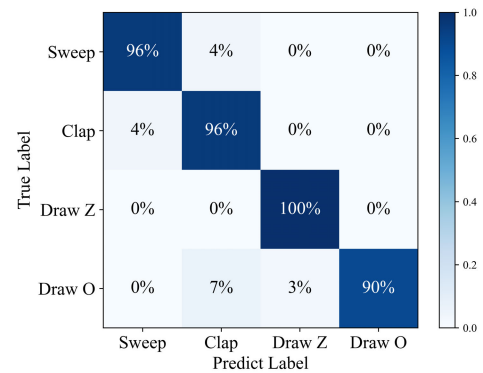


FIGURE 12. Confusion matrix for classification of 4 gestures by FEDRT-LSTM network model.

In conclusion, the highest accuracy for gesture recognition is achieved when the batch size is 32 and the learning rate is 0.001. The accuracy rate for gesture recognition reaches 95.54%, which is the highest among all 24 scenarios. Therefore, when using the CSI data collected by two receivers and the FEDRT-LSTM network model, the optimal choice for the loss function is MSE, the optimal epoch value is 200, the optimal learning rate value is 0.001, and the optimal batch size value is 32. Together, they form the optimal combination of hyperparameters.

At the same time, the confusion matrix of classifying 4 gestures using the FEDRT-LSTM network model is shown in Figure 12. Among them, the accuracy of recognizing the gesture of Draw Z is the highest, reaching 100%, while the accuracy of recognizing the gestures of Sweep and Clap is 96%, and the accuracy of recognizing the gesture of Draw O is the lowest, at 90%. This shows that FEDRT-LSTM can achieve a high recognition rate for the 4 in-vehicle CSI gesture data collected by two receivers.

V. EXPERIMENT RESULTS AND COMPARISON

A. EVALUATION INDICATORS

To evaluate the performance of the model in recognizing gestures, we use the following indicators [39].

- 1) Accuracy: Accuracy indicates the proportion of all gesture samples that are correctly classified by the

model and is an important measure of the overall classification correctness of the gesture recognition model. As shown in Equation (1).

$$Accuracy = \frac{1}{N} \sum_{i=1}^N I(y_i = \hat{f}(x_i)) \quad (1)$$

where N is the test sample size, $\hat{f}(X)$ is the learned model, and I is the indication function, which equals 1 when $y = \hat{f}(x)$, and 0 otherwise.

- 2) Precision: Precision is a measure of the proportion of all times that the model is predicted to be a certain gesture that it actually is that gesture. In other words, Precision describes the predictive accuracy of the model, i.e. the ratio of the number of samples correctly predicted by the model to be a certain gesture to the total number of samples predicted to be that gesture. As shown in Equation (2).

$$Precision = \frac{T_p}{T_p + F_p} \quad (2)$$

where T_p represents the number of true positives, F_p represents the number of false positives.

- 3) Recall: Recall is a measure of the proportion of all samples that are true for a certain gesture that the model successfully detects as that gesture. Recall reflects the model's ability to recognize a certain gesture, the ratio of the number of samples successfully recognized by the model as that gesture out of all the samples that are true for a certain gesture to the total number of samples that are true for that gesture. As shown in Equation (3).

$$Recall = \frac{T_p}{T_p + F_N} \quad (3)$$

where F_N represents the number of false negatives.

- 4) F1 Score: The F1 Score is the reconciled mean of Precision and Recall, which combines Precision and Recall to evaluate the overall performance of the model. It takes values between 0 and 1, with higher values indicating better model performance. As shown in Equation (4).

$$F1Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

The evaluation of these indicators gives us a more comprehensive overview of the model's performance in recognizing the gestures of Sweep, Clap, Draw Z, and Draw O, and fair comparison with other classification models.

To verify the performance of the designed FEDRT-LSTM network model on in-vehicle CSI gesture recognition collected by two receivers, DNN, GRU-CapNet, GRU, and LeNet-5 are selected for comparison. When all other models (DNN, GRU-CapNet, GRU, and LeNet-5) use the same optimal parameters for the FEDRTLSTM model, their accuracies are lower than that of FEDRTLSTM, as shown in Table 1, in decreasing order of accuracy.

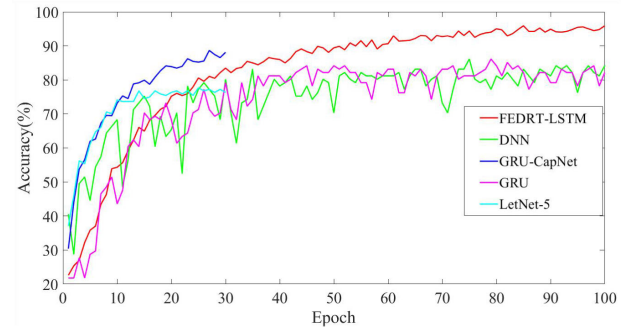


FIGURE 13. The graphs illustrate the accuracy and epoch curves for 5 different models when utilizing optimal hyperparameters.

B. COMPARISON

The selected evaluation indicators including Accuracy, Precision, Recall, and F1 Score of all 5 network models with optimized hyperparameters are compared in Table 2. FEDRT-LSTM with optimized hyperparameters achieves the highest scores in all indicators. DNN ranks the second, followed by GRU-CapNet, GRU, and LeNet-5. Therefore, the FEDRT-LSTM model performs best on the dataset used in this paper when the loss function is selected as MSE, epoch 200, the learning rate 0.001, and the batch size 32.

The accuracy and epoch curves for each of the 5 models using their respective optimal hyperparameter combinations are illustrated in Figure 13. It is worth noting that DNN, GRU-CapNet, GRU, and LeNet-5 all perform worse on in-vehicle CSI data for 4 gestures collected using two receivers than indoor dataset for 6 gestures, as expected, due to more complicated in-vehicle environment and fewer receivers. The proposed FEDRT-LSTM model demonstrates a significantly higher recognition accuracy than the other 4 network models.

Regarding the 5 network models, FEDRT-LSTM, DNN, GRU-CapNet, GRU, and LeNet-5, when using each optimal hyperparameters, the accuracy and epoch curves exhibit the following characteristics:

- 1) DNN: For small epoch values, the accuracy curve exhibits larger fluctuations. As the number of epochs increases, the accuracy gradually stabilizes and improves. However, compared to other complex network structures, DNN may require more epochs to reach higher levels of accuracy.
- 2) GRU-CapNet: GRU-CapNet combines the characteristics of GRU and Capsule Network. The accuracy v.s. epoch curve for GRU-CapNet show relatively faster improvements and convergence. As the number of epochs increases, the accuracy gradually improves and approaches stability.
- 3) GRU: In the initial stages, the accuracy curve for GRU may exhibit larger fluctuations. As the number of epochs increases, the accuracy gradually stabilizes and improves. However, similar to other complex model structures, GRU may require more epochs to achieve higher levels of accuracy.

TABLE 1. Results of accuracy when all other models (DNN, GRU-CapNet, GRU, and LeNet-5) use the same parameters as the FEDRT-LSTM model.

Network	FEDRT-LSTM	DNN	GRU-CapNet	GRU	LeNet-5
Loss Function	MSE	MSE	MSE	MSE	MSE
Epoch	200	200	200	200	200
Learning Rate	0.001	0.001	0.001	0.001	0.001
Batch Size	32	32	32	32	32
Accuracy	0.9554	0.8839	0.7893	0.6161	0.5982

TABLE 2. Comparison of optimal hyperparameters and evaluation indicators for 5 models.

Network	FEDRT-LSTM	DNN	GRU-CapNet	GRU	LeNet-5
Loss Function	MSE	CCE	BCE	MAE	MSE
Epoch	200	250	30	100	30
Learning Rate	0.001	0.001	0.001	0.005	0.005
Batch Size	32	32	16	16	32
Accuracy	0.9554	0.8839	0.8482	0.8393	0.7768
Precision	0.9568	0.8927	0.8499	0.8362	0.8065
Recall	0.9568	0.8838	0.8482	0.8373	0.771
F1 Score	0.9558	0.8836	0.8481	0.8367	0.7747

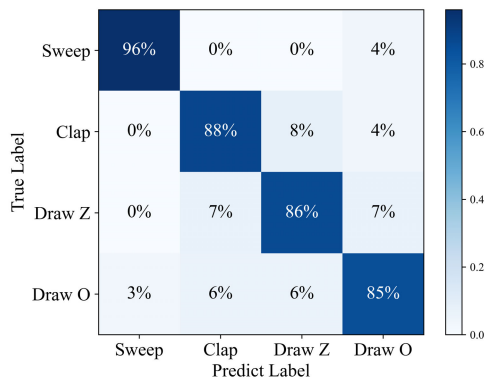


FIGURE 14. Confusion matrix for classification of 4 gestures by DNN.

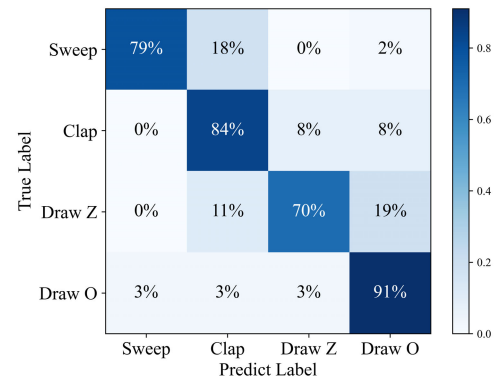


FIGURE 16. Confusion matrix for classification of 4 gestures by GRU.

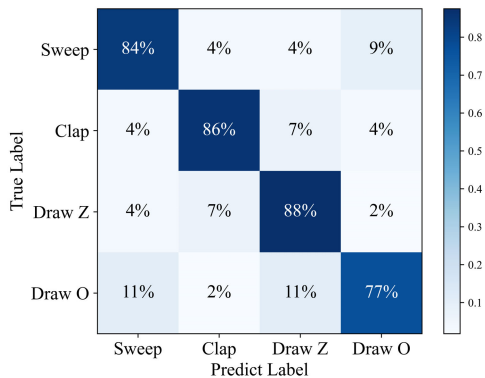


FIGURE 15. Confusion matrix for classification of 4 gestures by GRU-CapNet.

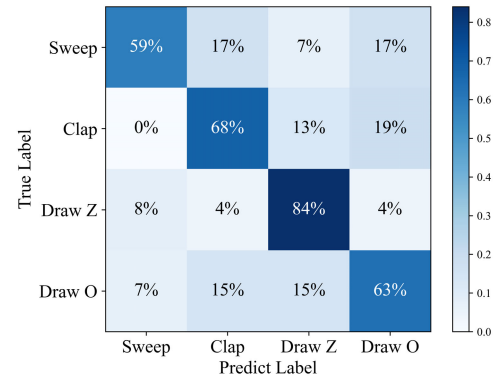


FIGURE 17. Confusion matrix for classification of 4 gestures by LetNet-5.

4) LeNet-5: LeNet-5 is a classic convolutional neural network architecture. The accuracy and epoch curve for LeNet-5 demonstrates a gradually increasing trend that tends to stabilize. In the initial stages, there may be significant improvements in accuracy. As the number of epochs increases, the rate of improvement may gradually slow down, but there is still potential for further accuracy enhancement.

5) FEDRT-LSTM: Initially, the accuracy curve may experience significant fluctuations as the model needs to adapt to the long-term dependencies in the sequence data. As the number of epochs increases, the accuracy gradually stabilizes and improves, forming a curve that shows a gradual ascent and convergence.

For the DNN model using the optimized hyperparameters, its classification confusion matrix for the 4 gestures is shown in Figure 14. Among them, the recognition accuracy of Sweep

gestures is the highest at 96%, Clap hand is 88%, Draw Z is 86%, and Draw O is the lowest at 85%.

For the GRU-CapNet model using the optimized hyperparameters, its classification confusion matrix for the 4 gestures is shown in Figure 15. It can be observed that the highest recognition accuracy of 88% is achieved for the Draw Z gesture, 86% for the gestures of Clap, 84% for the gestures of Sweep, and 77% for Draw O.

For the GRU model using the optimized hyperparameters, its classification confusion matrix for the 4 gestures is shown in Figure 16. It can be observed that the highest recognition accuracy of 91% is achieved for the Draw O gesture, 84% for the Clap, 79% for the Sweep, and 70% for the Draw Z gesture.

For the LeNet-5 model using the optimized hyperparameters, its classification confusion matrix for the 4 gestures is shown in Figure 17. It can be observed that the highest recognition accuracy of 84% is achieved for the Draw Z gesture, 68% for the Clap, 63% for the Draw O, and 59% for the Sweep gesture.

VI. CONCLUSION

The experiment results on the in-vehicle CSI gesture dataset collected by the two receivers show that the proposed FEDRT-LSTM with optimized hyperparameters outperforms the other 4 models, which confirms the effectiveness and reliability of the method proposed in this paper.

However, considering the complexity of the in-vehicle electromagnetic environment, we realize that the model faces new challenges. For example, the layout of different vehicles, the number of passengers, and different types of vehicles (such as electric vehicles) may all have an impact on the electromagnetic environment, posing challenges to the applicability of the model. We need to further explore whether the model needs to be retrained under these variables to ensure its robustness and accuracy in different situations. These challenges are of great significance for the application of this technology in vehicles. Therefore, future research will not only focus on in-vehicle CSI gesture recognition for single receiver but also explore the generalization of the FEDRT-LSTM model in different electromagnetic environments, aiming to provide more practical guidance for the industrial application of the model.

Specifically, study of a single receiver can directly use the dataset in this paper, and the BVP features can be obtained by processing the CSI signals of a single receiver. When verifying the generalization ability of the model in different electromagnetic environments, we may use the self-built in vehicle gesture dataset with 3 Wi-Fi receivers using GRU CapNet in [18] as the test set, which is different from our dataset in this paper in terms of vehicle type, experimental layout, number of receivers, and gesture categories.

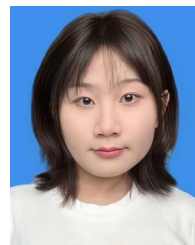
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